

ECE LAB ASSISTANT

AI-Assisted Web Application for Electronics and Communication Engineering
Students

Mini Project Documentation

Project Type: B.Tech Mini Project

Domain: Electronics and Communication Engineering (ECE)

Development Platform: Base44 AI

Live Application: <https://nebulous-circuit-pilot-lab.base44.app/>

Academic Year: 2026–2027

1. Abstract

ECE Lab Assistant is a web-based educational application designed to support B.Tech Electronics and Communication Engineering students during laboratory learning and preparation. The application brings useful ECE learning resources into a single platform, including electronics calculations, laboratory information, component references, formulas, and viva preparation resources.

The project was developed using Base44 AI, an AI-assisted application development platform that enables applications to be created and refined using natural-language requirements. The project demonstrates how AI-assisted development can be applied to create a practical educational tool for engineering students.

2. Objectives

- Provide a centralized platform for ECE laboratory learning.
- Help students perform common electronics calculations.
- Provide quick access to electronics formulas.
- Provide information about common electronic components.
- Assist students in understanding laboratory experiments.
- Support viva-voce preparation.
- Provide an easy-to-use and responsive web interface.
- Demonstrate the practical use of AI-assisted application development.

3. Problem Statement

ECE students often need to refer to different sources while preparing for laboratory sessions. They may need separate resources for circuit calculations, electronics formulas, component information, laboratory procedures, experiment preparation, and viva questions. Switching between multiple resources can make preparation inconvenient.

ECE Lab Assistant addresses this problem by bringing these learning resources together in one web application.

4. Proposed Solution

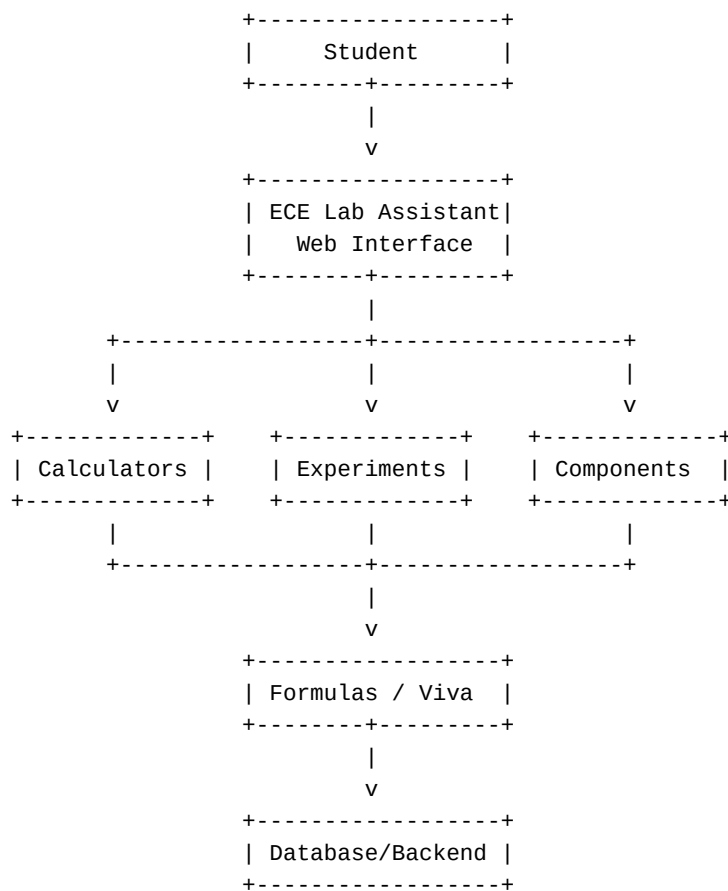
ECE Lab Assistant provides a centralized digital platform where students can access important ECE laboratory resources. The application is organized around electronics calculators, laboratory experiments, electronic component references, formula resources, viva preparation, and a student dashboard.

Live Demo: <https://nebulous-circuit-pilot-lab.base44.app/>

5. Technology Stack

Technology / Concept	Purpose
Base44 AI	AI-assisted web application development and deployment
Web Application	Browser-based student learning platform
Responsive UI/UX	Support for desktop, tablet and mobile screens
Database	Store application and student-related data where configured
Authentication	Student/admin access control where configured
Mathematical Logic	Electronics calculations and formula-based tools

6. System Architecture



7. Major Modules

7.1 Dashboard

The dashboard provides a central starting point for students and gives quick access to the application's major learning resources and activities.

7.2 Electronics Calculators

The calculator section is designed to perform common electronics calculations and reduce manual calculation effort.

- **Ohm's Law:** $V = I \times R$
- **Power:** $P = V \times I$, $P = I^2 R$, $P = V^2 / R$
- **Voltage Divider:** $V_{out} = V_{in} \times R_2 / (R_1 + R_2)$
- **RC Time Constant:** $\tau = R \times C$
- **RL Time Constant:** $\tau = L / R$

7.3 Laboratory Experiments

The experiment section organizes laboratory learning material. An experiment page can contain the aim, components required, theory, circuit information, procedure, formula, observation, result, precautions, and viva questions.

Example experiments include Ohm's Law, Kirchhoff's Laws, Series and Parallel Circuits, Diode Characteristics, Zener Diode Characteristics, Transistor Characteristics, Rectifiers, Operational Amplifiers, Logic Gates, and 555 Timer.

7.4 Electronic Component Library

The component section provides reference information about commonly used electronic components such as resistors, capacitors, inductors, diodes, LEDs, Zener diodes, transistors, MOSFETs, operational amplifiers, logic ICs, sensors, relays, and transformers.

7.5 Formula Library

The formula library organizes important ECE formulas into categories such as Circuit Theory, Network Theory, Analog Electronics, Digital Electronics, Signals and Systems, Communication Systems, Control Systems, Electromagnetics, and Microprocessors.

7.6 Viva Preparation

The viva section supports preparation for laboratory viva examinations through topic-wise questions, answers, multiple-choice questions, random questions, quiz scoring, explanations, and performance tracking.

8. Student Features

- Student registration/login where configured.
- Personal dashboard.
- Experiment tracking.
- Saved calculations.
- Formula reference.
- Component reference.
- Viva practice.
- Quiz performance tracking.

9. Database and Data Management

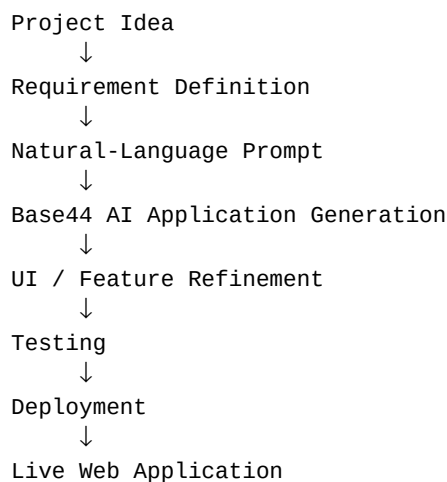
The application can organize information around students, experiments, experiment records, components, calculations, formulas, viva questions, and quiz attempts. Base44 provides database and application-building capabilities that can be used to persist application data where configured.

10. User Interface and UX

- Clean and professional engineering-oriented interface.
- Responsive layout for desktop, tablet, and mobile.
- Simple navigation and clear information hierarchy.
- Cards, dashboards, search and filtering.
- Readable typography and consistent navigation.
- Student-friendly interaction flow.

11. Role of AI in Development

A key aspect of this project is AI-assisted development. Requirements were expressed through natural-language prompts and used to generate and refine the application.



12. Development Methodology

Step 1 — Identify the Problem

Identify common difficulties faced by ECE students during laboratory preparation.

Step 2 — Define Requirements

Determine the required modules such as calculators, experiments, components, formulas and viva preparation.

Step 3 — Create the Initial Prompt

Provide the application requirements to Base44 AI in natural language.

Step 4 — Generate the Application

Generate the initial application structure and functionality.

Step 5 — Refine the Application

Improve UI design, navigation, content, features and user experience through additional prompts.

Step 6 — Test

Test the major screens, navigation, calculations and expected workflows.

Step 7 — Deploy

Publish the application as a web application.

Step 8 — Document

Prepare project documentation and GitHub materials.

13. Testing

Test Area	Expected Result
Application loading	Application loads successfully
Navigation	Major pages open correctly
Calculator inputs	Valid inputs produce appropriate calculations
Invalid inputs	Validation or error feedback is displayed
Experiment pages	Experiment information is displayed correctly
Component search	Relevant components can be located
Formula section	Formula information is displayed
Viva section	Questions and answers work as designed
Responsive design	Interface adapts to different screen sizes
Authentication	Authorized users access appropriate features

14. Advantages

- Centralized ECE laboratory resources.
- Easy-to-use interface.
- Reduces manual calculation effort.
- Helps students prepare for laboratory sessions.
- Provides quick formula references.
- Supports viva preparation.
- Accessible through a web browser.

- Responsive design.
- Demonstrates practical AI-assisted software development.
- Can be extended with additional ECE features.

15. Limitations

- The application does not replace physical laboratory equipment.
- Simulated calculations do not represent every real-world circuit condition.
- Component information should be verified against manufacturer datasheets for practical hardware work.
- AI-generated functionality depends on the quality and clarity of the requirements and prompts.
- Advanced circuit simulation may require specialized tools.

16. Future Enhancements

- **AI ECE Tutor:** Add an AI assistant that explains ECE concepts and formulas.
- **Circuit Simulator:** Add interactive digital circuit simulation.
- **IoT Dashboard:** Connect real sensors and display live readings.
- **Advanced Calculators:** Add BJT, MOSFET, Op-Amp, filter and communication-system calculators.
- **Progress Analytics:** Add charts for experiment completion and quiz performance.
- **Faculty Dashboard:** Allow faculty to upload experiments, questions and monitor student progress.
- **Mobile Application:** Extend the web application into a dedicated mobile experience.

17. Expected Outcomes

- A functional ECE learning web application.
- Centralized laboratory resources.
- Interactive electronics calculations.
- Organized experiment information.
- Component and formula references.
- Viva preparation resources.
- Responsive user interface.
- Practical experience with AI-assisted application development.

18. Suggested GitHub Repository Structure

ECE-Lab-Assistant/



```
■■■ README.md
■■■ docs/
■   ■■■ project-report.md
■   ■■■ screenshots/
■       ■■■ home.png
■       ■■■ dashboard.png
■       ■■■ calculators.png
■       ■■■ experiments.png
■       ■■■ components.png
■       ■■■ viva.png
■■■ prompts/
■   ■■■ base44-prompts.md
■■■ LICENSE
```

19. Project Team

Project Title: ECE Lab Assistant

Department: Electronics and Communication Engineering

Project Type: Mini Project

Development Platform: Base44 AI

Team Members: [Student Name] — [Roll Number], [Student Name] — [Roll Number], [Student Name] — [Roll Number]

Institution: [College Name]

Academic Year: 2026–2027

20. Project Guide

Guide Name: [Faculty Name]

Designation: [Designation]

Department: Electronics and Communication Engineering

21. Conclusion

ECE Lab Assistant demonstrates the development of an educational web application using AI-assisted software development. The project combines ECE laboratory resources with a modern web-based interface to create a centralized learning platform for students.

The project provides a foundation for future improvements such as AI tutoring, circuit simulation, IoT integration, advanced analytics, and faculty management.

22. References

1. Base44 — AI App Builder: <https://base44.com/ai-app-builder>
2. Base44 — Features: <https://base44.com/features>
3. Base44 — Developer Platform: <https://base44.com/developers>

4. ECE Lab Assistant — Live Application: <https://nebulous-circuit-pilot-lab.base44.app/>